Moontaha Farin

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Summary

- Biomedical Engineer (MS) with research experience in **soft polymers, polymer synthesis, DLP 3D printing, drug delivery, and polymer characterization**, including NMR, FTIR, XRD, DSC, DMA, tensile testing, SEM, GPC, and UV–Vis spectroscopy.
- Experienced in **biocompatibility evaluation**, including live/dead assay, cytotoxicity, antibacterial testing, hemolysis assays, histology, and in vivo studies.
- Published researcher in **Rapid Prototyping Journal** and recipient of an **NIH R13 Travel Award**
- **U.S. Lawful Permanent Resident (Green Card holder); no visa sponsorship required.**

Education

University of Akron Aug 2024 - Aug 2026

MS in Biomedical Engineering

- **Resaerch Project:** Synthesis and characterization of a photocrosslinkable shape memory polymer for biomedical applications
- **Notable Courses:** Model Based System Engineering, 3D Bioprinting, Advanced Polymer Engineering, & Experimental Design

Bangladesh University of Engineering & Technology Feb 2015 - Apr 2019

MS in Biomedical Engineering

- **Thesis:** Physically modified sweet potato starch/gelatin based folic acid orally disintegrating film for rapid drug release
- **Notable Courses:**Advanced Biomaterials, Regenrative Medicine & Artificial Organ, Translational Neural Engineering, Chemistry of Polymers, Advanced Drug Delivery, & Advanced Biophysics

Bangladesh University of Engineering & Technology Feb 2015 - Apr 2019

BS in Chemical Engineering

- **Thesis:** Design optimization for synthesis of activated carbon fiber from textile cotton waste

Skills

Lab skills: Material Characterization (NMR, XRD, DSC, FTIR, UV-Vis Spectroscopy,SEM, GPC), 3D printing, Electrospinning, Drug dissolution test, Antibacterial test, Cell Culture, Hemolysis assay, Histology, Animal Handling, In vivo test development for wound healing

Software skills: ImageJ, OriginLab,LaTeX Solidworks, Simulink, Tina-Ti, MATLAB, Arduino, Biorender, ChemDraw

Experience

Graduate Research Assistant Aug 2024 – Aug 2026

University of Akron, Department of Biomedical Engineering *Akron, OH*

- Developed and characterized a novel photocrosslinkable shape memory polymer for biomedical application
- Conducted chemical, thermal, mechanical characterizations such as **NMR, FTIR, XRD, DSC, DMA & tensile test**
- Performed cytoxicity test using **live/dead assay**

Teaching Assistant Jan 2025 – May 2025

University of Akron *Akron, OH*

- Evaluated exam papers for BMEN 4701: Human Factors Engineering 001 & BMEN 362-001: Transport Fundamentals for Biomedical Engineering

- Guided undergraduate students in applying transport phenomena concepts to problem solving and analysis

Graduate Researcher April 2021 – July 2024
Bioinnovation Research Group, Department of Biomedical Engineering *Dhaka, Bangladesh*

- Developed a drug delivery platform using oral disintegrating films to avoid first-pass metabolism of therapeutic molecules.
- Controlled functional properties of polymeric matrices such as thin films.
- Performed XRD, FTIR, SEM, DSC,hemolysis assays, antibacterial tests, and histological analyses for biomaterial safety and functionality.

Teaching Assistant Jan 2022 – May 2023
Bangladesh University of Engineering & Technology *Dhaka, Bangladesh*

- Guided 50+ students through hands-on biomaterials and drug delivery lab
- Trained students on **SOP adherence**, equipment troubleshooting, and good laboratory practices (GLP), assisted in developing lab manuals

Undergraduate Researcher Oct 2019 – June 2020
Bangladesh University of Engineering & Technology *Dhaka, Bangladesh*

- Studied the preparation and adsorption performance of activated carbon fiber from textile cotton wastes

Publications

Additive Manufacturing: A Bespoke Solution for Personalised Medicine
Authors: Moontaha Farin, JT Mysha, I. Gibson, MT Arafat *Rapid Prototyping Journal*

- Available at: https://doi.org/10.1108/RPJ-09-2023-0326

Role of Physical Modification of Starch on Tuning the Properties of ODF *Submitted*
Authors: Moontaha Farin, SY Wong, Xu Li, SC Bachar, MT Arafat

Oral Presentations

- **Society For Biomaterials (SFB) 2026 Annual Meeting & Exposition**, Atlanta, GA — Oral Presentation: *"Synthesis and Characterization of Poly(Glycerol Dodecanedioate) Methacrylate as a Photocrosslinkable Shape Memory Biomaterial."*

Poster Presentations

TERMIS-AM 2025 2025
Poster presentation

- Presented a research poster; received an NIH R13 travel award to support attendance and presentation.

Awards
Achievements

NIH R13 travel award Got NIH R13 travel award to present poster on TERMIS-AM 2025 conference

Hult Prize, Kuala Lumpur, Malaysia My team's social startup, TurnAbout, was established to provide biogas in rural areas using FTG bags (food-turned-gas bags). The concept was promoted at our own university, and out of 54 countries, we placed among the top 6 in the Kuala Lumpur region (Jul 2018)

Best Poster Presented a poster at the International Symposium on Chemical Safety and Security Management 2020, organised by the UN on the reactive hazards of substandard hydrogen gas balloon cylinders (Jan 2020), Available at: http://dx.doi.org/10.13140/RG.2.2.17932.90249

Projects

Design and Evaluation of a Microfluidic Passive Mixer — Designed and evaluated a 3D-printed serpentine-channel microfluidic passive mixer (*Model, Report, and Presentation*).

Engineering Islet Encapsulation Device for Islet Transplantation in Type-1 Diabetes — Reviewed and critiqued literature on islet transplantation with focus on vascularization and nutrient delivery (*Report and Presentation*).

Hospital-Based Medical Device: Identifying Needs — Collected data from dentistry and ENT units and collaborated with clinicians to develop methods for reducing medical device failure rates (*Report and Presentation*).

Design of a Low-Cost Rainwater Filter Using PVA Electrospun Fiber — Developed an integrated rainwater filter with PVA electrospun fiber for rural areas in Bangladesh (*Research and Development*).

Training

Training on Conduct of Animal Study according to Ethical Guideline of ICDDR,B — Learned about handling, blood withdrawal, feeding, breeding, and injection routes in mice, rats, guinea pigs, and rabbits.

Training on Radiation and Safety for radiation-generating equipment by University of Akron-Trained on the safety and hazard prevention of using XRay, UV IR generating equipment .

Industrial Attachment for Chemical Engineering at Sylhet Gas Field — Monitored gas production and process safety regulations in Sylhet Gas Fields Company Limited.

Leadership Experience

American Institute of Chemical Engineering BUET Chapter |Secretary, Fund Raising| Managed budget, raised funds and organized outreach activities (Jul 2022)

United Nations Industrial Development Organization (UNIDO)|Volunteer| Volunteered on the seminar on "Efficient Energy Management for Resource Efficiency and Cleaner Production" organized by UNIDO (Jan 2018)